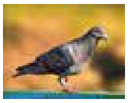

One pigeon to rule them all
Of the 290 pigeon species in existence, only one, the rock dove has adapted to urban life **https://dgit.in/may19-79**

The Instagram look is dead
Influencers are ditching the manufactured glossy aesthetic for images that are more raw and natural **https://dgit.in/may19-80**

FAMOUS EQUATIONS

These are the equations that not just scientists and mathematicians, but everyone should know about considering their far reaching implications.

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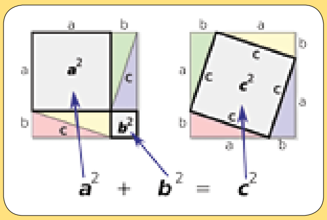
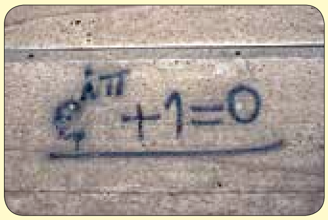
Equations are hard to understand because of the strange notations and relationships. They are often described in technical language, and look abstract on paper. Still, these equations dictate nearly all aspects of our daily existence. While the math behind them might be difficult to understand, their implications and relevance to us is not. The mapping application on your smartphone for example, directly uses three of these equations every time it is run. The more complex equations towards the bottom of the list, are derived from the simpler equations towards the top of the list. These are the equations that form the basis of civilisation and technological advancement.



PYTHAGORAS' THEOREM

This is perhaps the simplest equation on the list. It relates the size of the hypotenuse of a right-angled triangle to the other two sides – the square of the hypotenuse is equal to the sum of the squares of the other two sides. Although widely attributed to Pythagoras of Samos, a mathematician from Ionia, there are instances of it being used in Egypt, Babylon, India and China hundreds of years before Pythagoras was born. Over the years, it has been proved many times, through various approaches. The earliest practical application was to construct buildings. Cartographers also have to thank this theorem, as it allows them to triangulate accurately. The same principles are carried over to space-based applications, and GPS technologies also use triangulation to pinpoint a device on the map. If you use any location based apps to call for cabs or food, then it is because of this theorem.



EULER'S IDENTITY

Euler's identity is called the most beautiful equation in mathematics, and there are entire books that just explore its beauty and elegance. In a study conducted on the brains of mathematicians, the orbitofrontal cortex of the brain lit up when exposed to the equation. This is the part of the brain that gets activated when exposed to aesthetic music, poetry or pictures. The brilliance of the equation lies in intimately tying together five fundamental mathematical constants: 0,1, pi, e and i. The base of the natural logarithm is e, while i is the imaginary unit. Edward Kasner and James R Newman in *Mathematics and the Imagination* explained, "it appeals equally to the mystic, the scientist, the philosopher, the mathematician". At the time of its discovery, it was known to be true as it had been proved. However, it was not well understood, and even considered paradoxical. Over time, though, mathematicians have fallen in love with it.

MAXWELL'S EQUATIONS

James Clerk Maxwell published a series of equations between 1861 and 1862, that were derived from the works of other scientists including Gauss, Faraday, Ampere and Ohm. While Maxwell originally came up with a set of twenty equations, Oliver Heaviside reduced them to only four. The implications of the equations were radical at that time, and predicted the existence of electromagnetic waves before they were discovered. At that time it was believed that light required a medium called the luminiferous aether to propagate, which was not a requirement according to the equations. Maxwell's equations together describe the fundamental laws of electricity, light and magnetism, and combine them together into a single theory of electromagnetism. Everything from electricity to the information age is thanks to Maxwell's equations. Computers, smartphones, television sets, GPS satellite constellations, toasters, refrigerators, washing machines and the entire electricity grid, are designed and realised based on Maxwell's equations.